

Pathao: Order Delivery Management System

CSE326: Information System Design Sessional

pathao.com

Group: 9

Roll Numbers: 2105039, 2105040, 2105045, 2105056, 2105057, 2105058

Project Overview: Pathao's Delivery Ecosystem

Market Leader

Bangladesh's top delivery platform, serving 10M+ users in 50+ cities.

High Volume

Processes over 100,000 parcels daily, demonstrating high operational capacity.

Streamlined Workflow

Features a streamlined five-stage delivery process for efficiency.



Feature 1: Order Creation & Management

- Merchants easily submit delivery orders with pickup and delivery details.
- Orders are validated in real-time and initially set to **PENDING** status.
- Merchants can view, update, or cancel orders before assignment.



Feature 2: Driver Assignment & Matching

Geospatial Query

Available drivers are identified by location.

Optimal Match

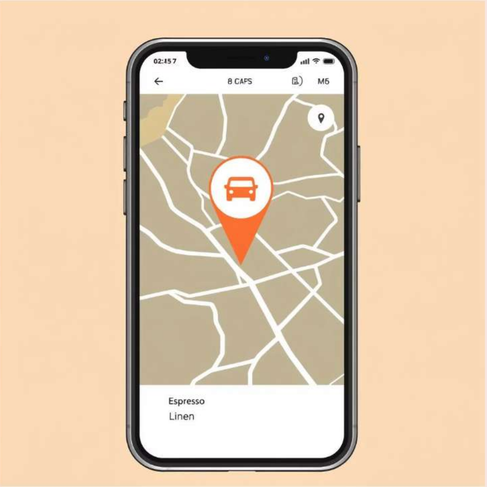
The nearest driver is matched to the pickup location.

Instant Assignment

Orders are assigned, and drivers receive notifications.

Feature 3: Real-Time GPS Tracking & Location Updates

- Drivers share real-time location data.
- Dynamic ETA based on traffic.
- Customers view live driver movement.
- See remaining distance and updated ETA.



Feature 4: Order Status & Notifications

01

Lifecycle Progression

Orders progress through key stages: Pending, Assigned, Picked Up, In Transit, Delivered.

02

Automated Alerts

Automated alerts instantly inform customers, merchants, and drivers of status changes.

03

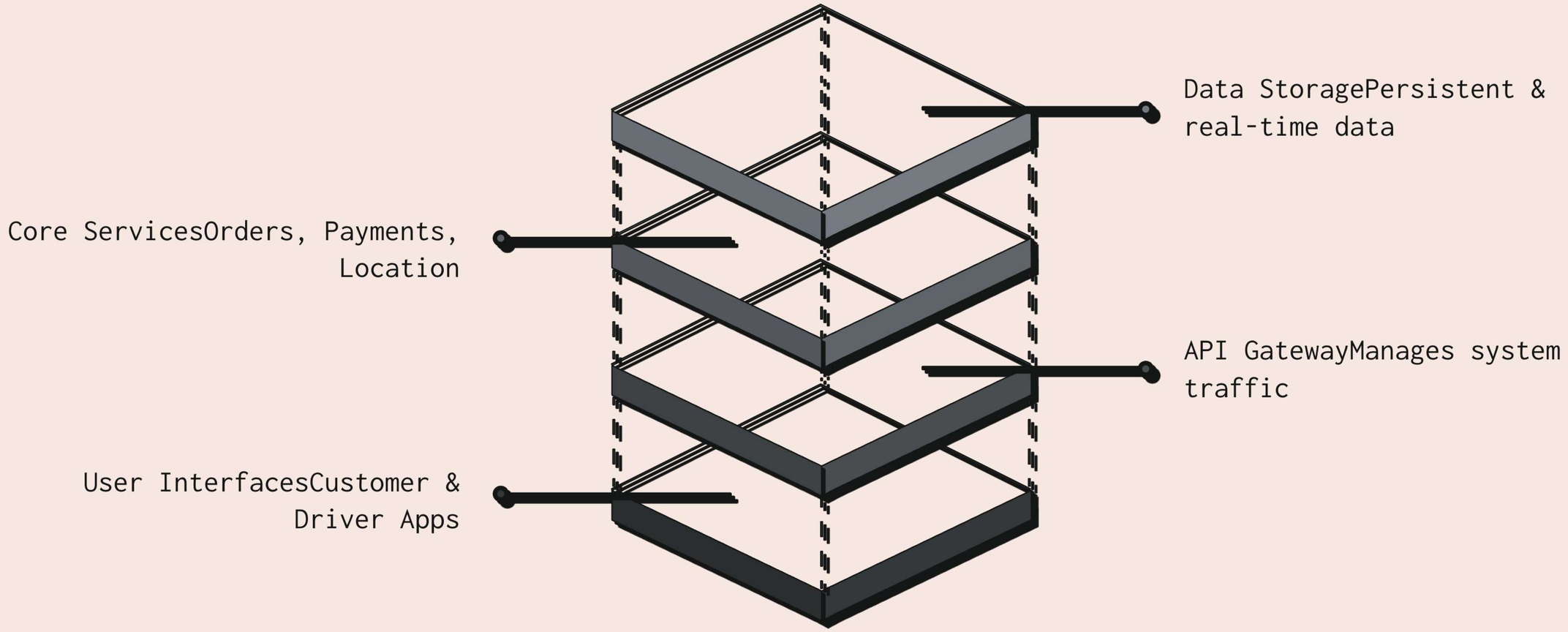
Delivery Verification

Deliveries are confirmed with photo proof or a one-time password for enhanced security.

Feature 5: Payment & Verification

- Supports Cash on Delivery (COD) and prepaid options.
- Drivers collect cash payments at delivery.
- Drivers confirm payment via app.
- Payment status recorded for reconciliation.

Technical Architecture & Scalability

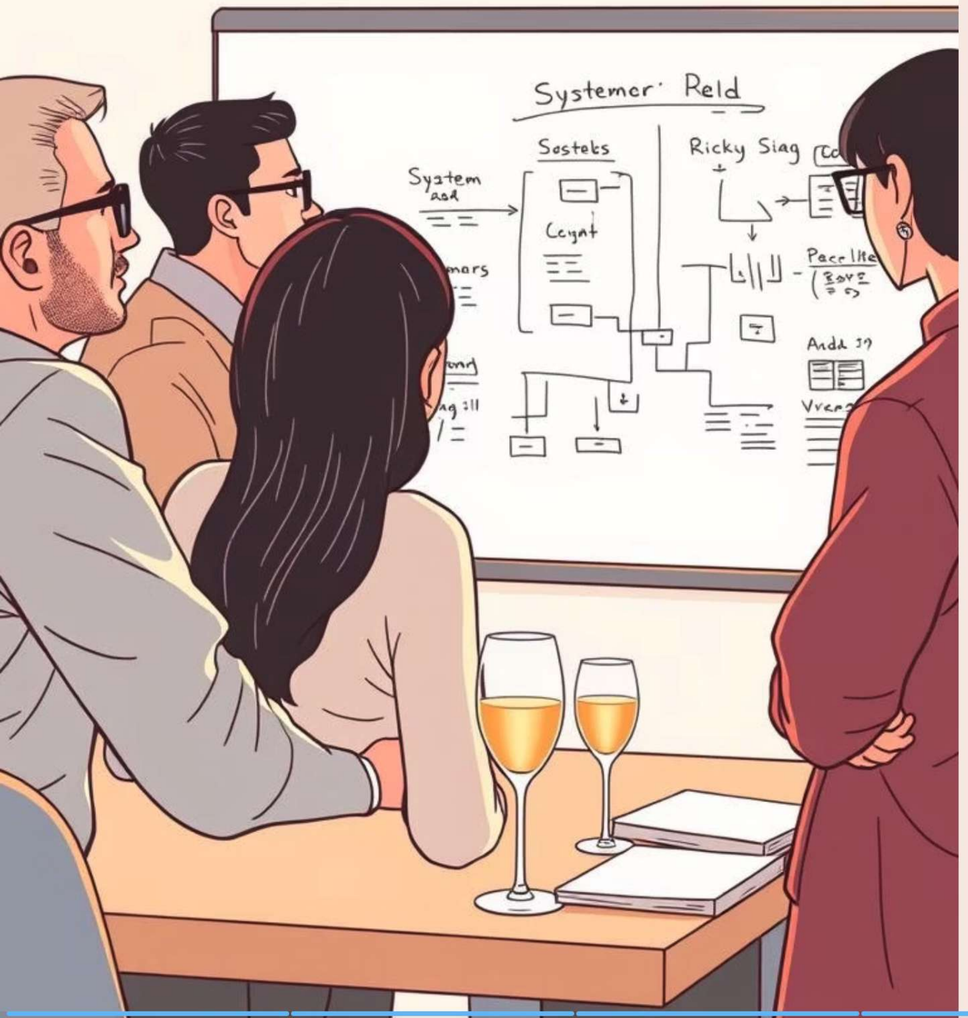


Core Technologies

- **Frontend:** Delivers dynamic and real-time user experiences.
- **Backend:** Powers high-performance API services for operations.
- **Database:** Stores data securely and enables fast access.

Key Challenges

- Location Updates
- Geospatial Queries
- Notification Delivery



Why This Module Matters: Impact & Learnings

Key Learning Outcomes

- Designing real-time systems effectively.
- Optimizing geospatial data handling.
- Ensuring database scalability.

Thank you!